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SGD

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GD

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SGD

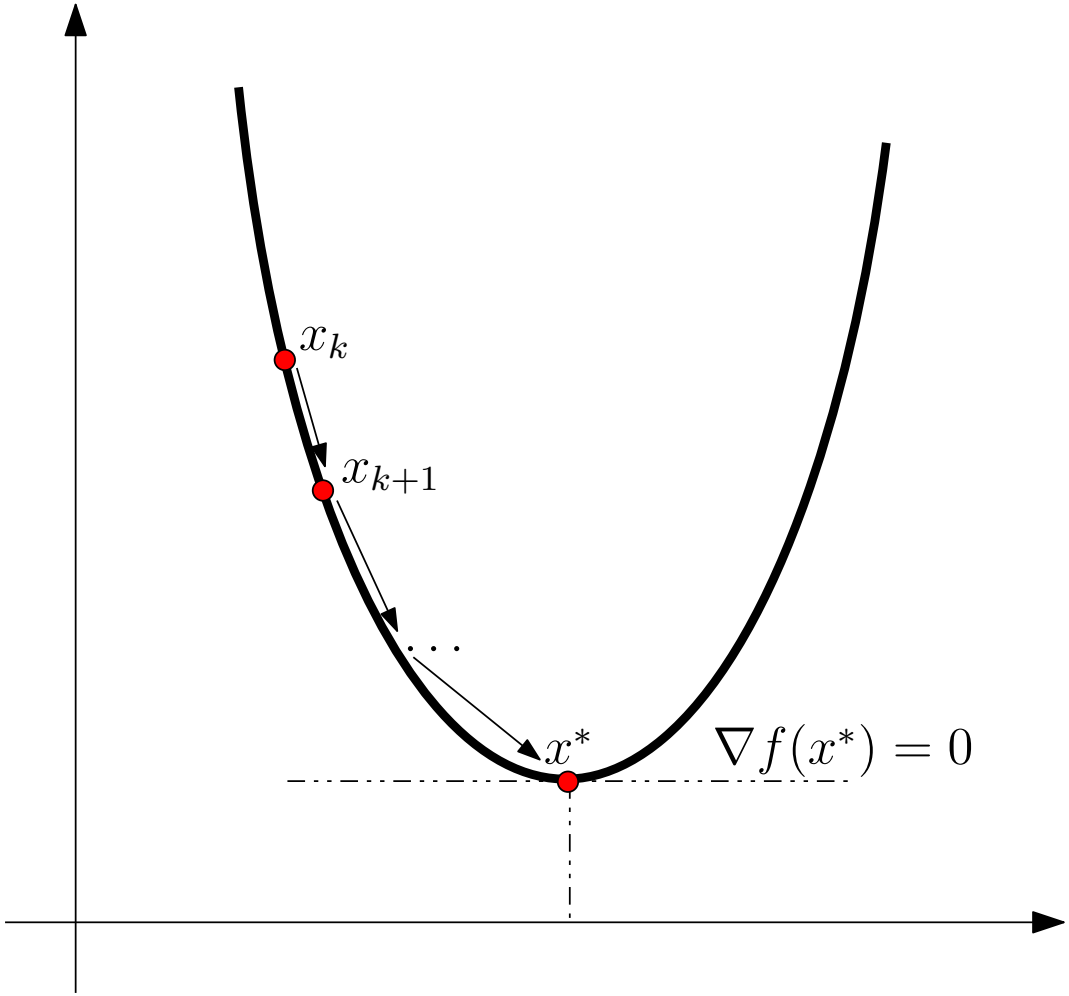
$g(x) \in \partial f(x)$

Subgrad

Gradient Descent

$$x \leftarrow x - \eta \nabla f(x)$$

Descent methods



Descent methods

- Suppose we have a vector $x \in \mathbb{R}^n$ for which $\nabla f(x) \neq 0$
- Consider updating x using

$$x(\eta) = x + \eta d,$$

where **direction** $d \in \mathbb{R}^n$ obtuse to $\nabla f(x)$, i.e.,

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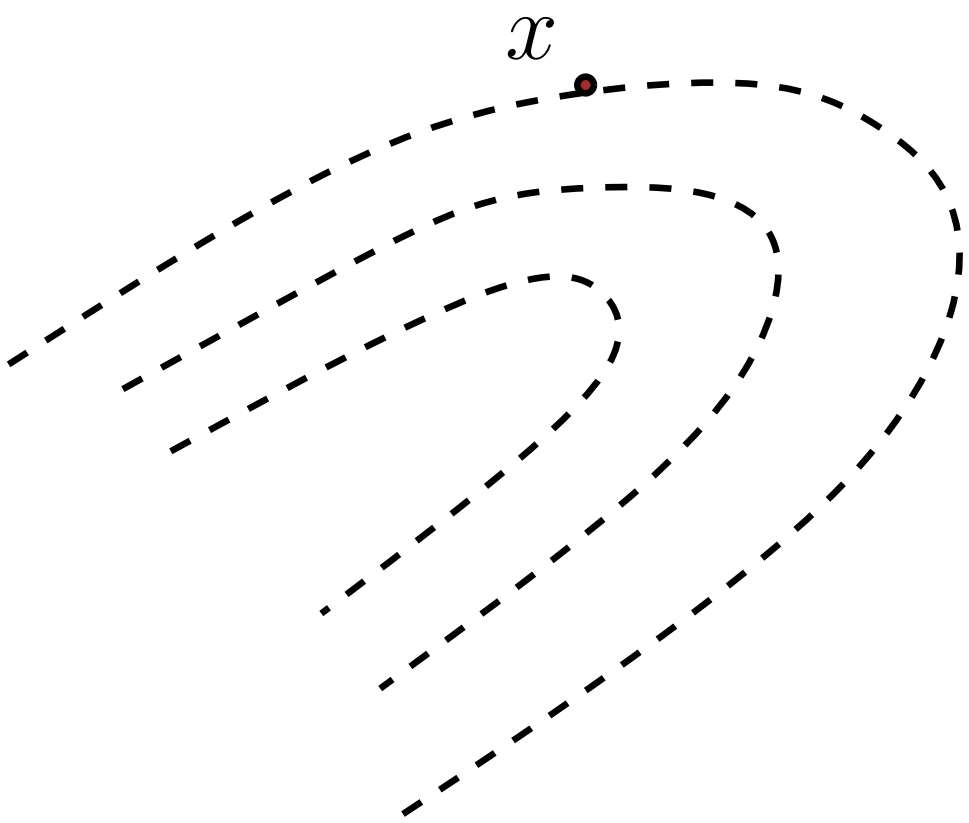
- Since d is obtuse to $\nabla f(x)$, this implies $f(x(\eta)) < f(x)$

Descent methods



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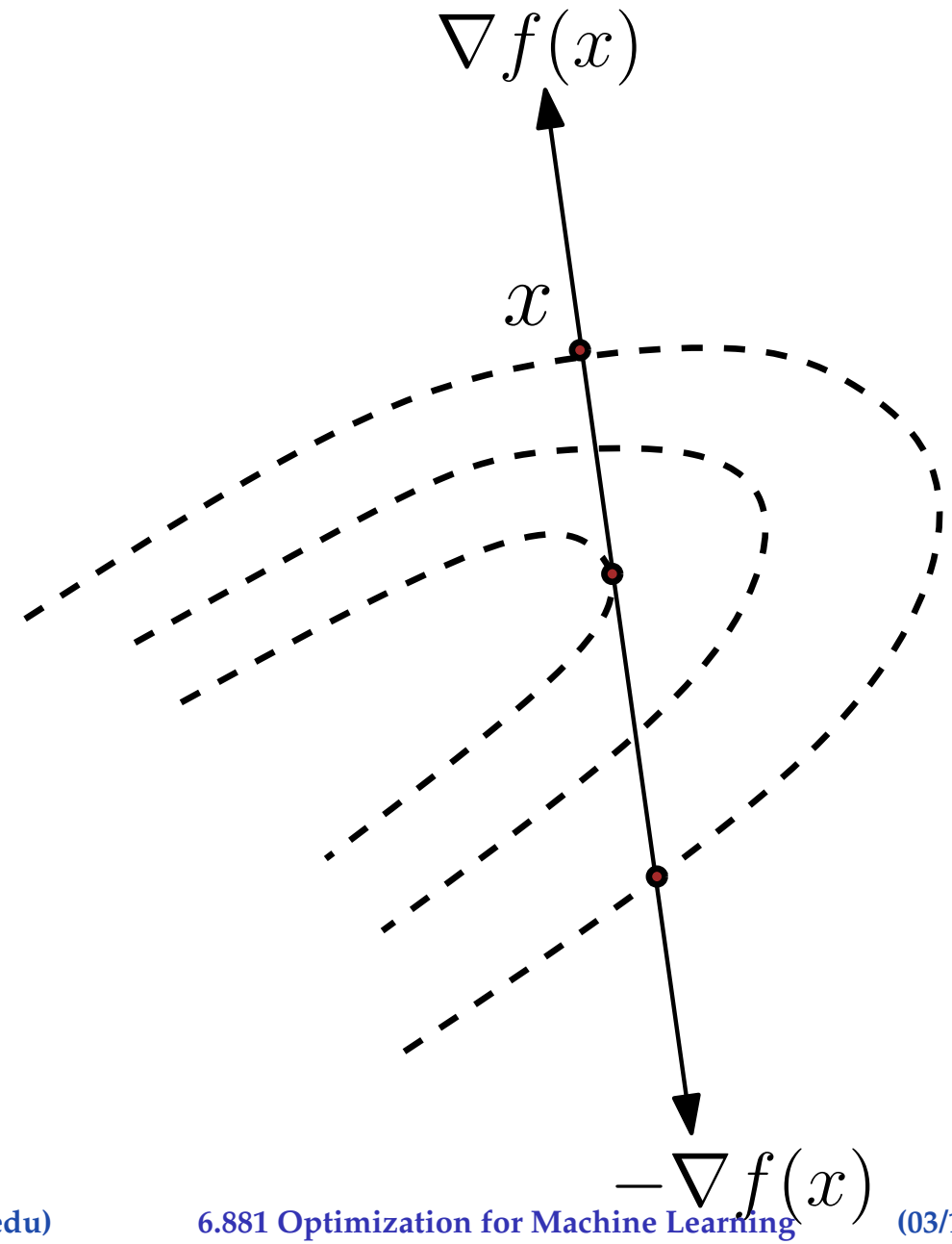

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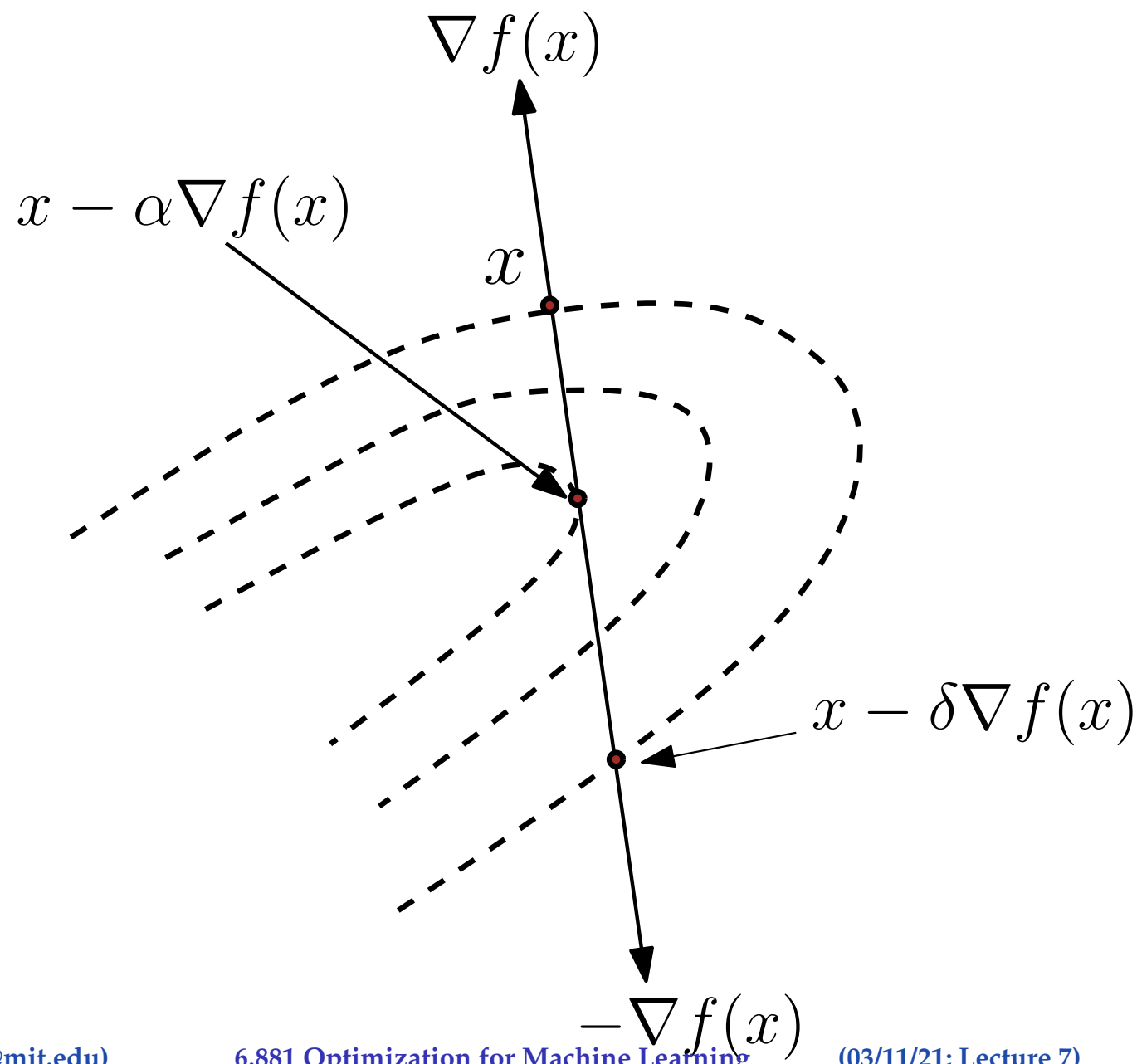

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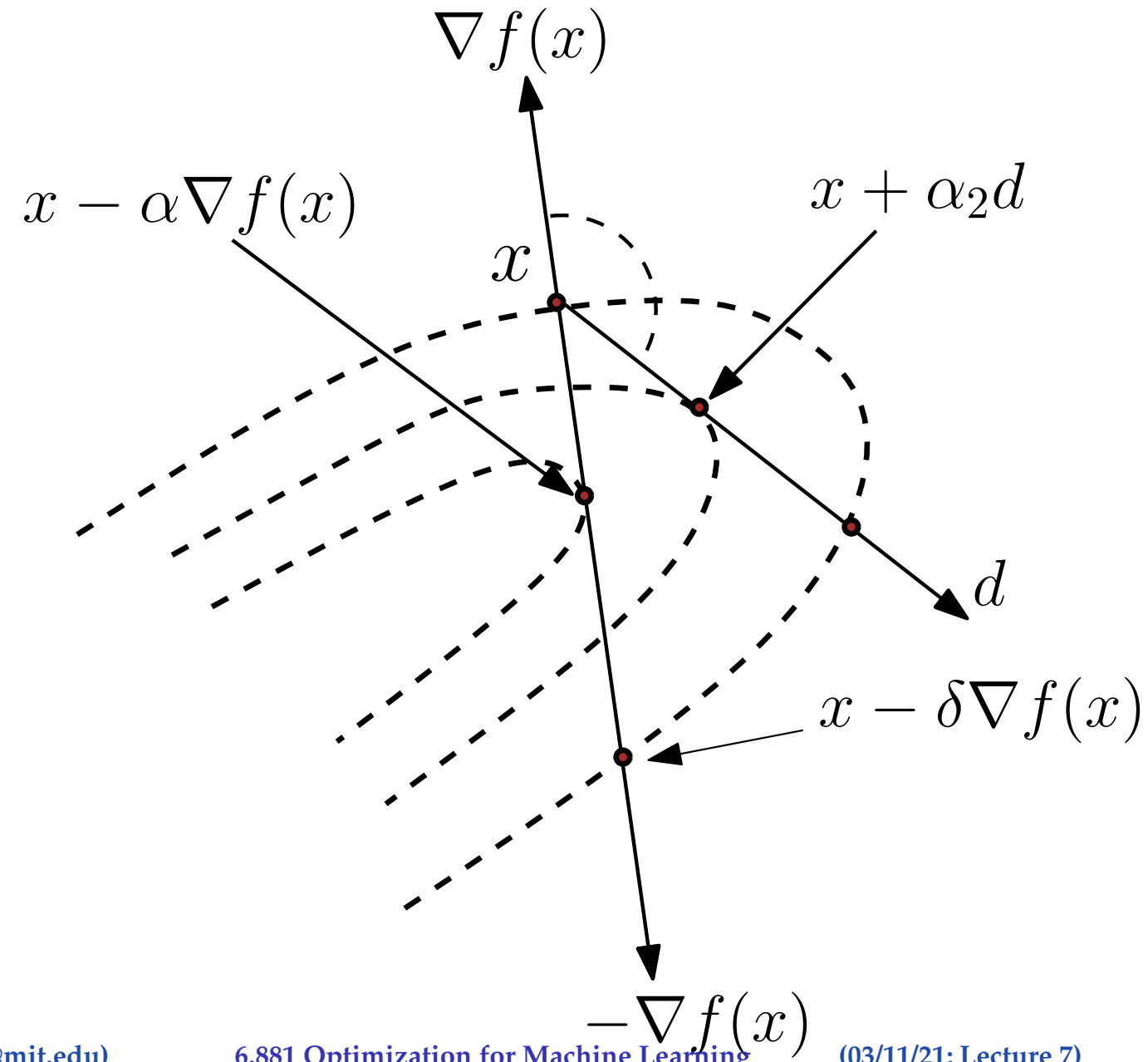

Gradient-based methods

1. Start with some guess x^0;
2. For each $k = 0, 1, \ldots$
   - $x^{k+1} \leftarrow x^k + \eta_k d^k$
   - Stop somehow (e.g., if $\|\nabla f(x^{k+1})\| \leq \varepsilon$)

Gradient based methods

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Numerous ways to select η_k and d^k

Many methods **seek monotonic descent**

$$f(x^{k+1}) < f(x^k)$$

Gradient methods – direction

$$x^{k+1} = x^k + \eta_k d^k, \quad k = 0, 1, \ldots$$

- Different choices of direction d^k
  - **Scaled gradient:** $d^k = -D^k \nabla f(x^k)$, $D^k \succ 0$
  - **Newton's method:** $(D^k = [\nabla^2 f(x^k)]^{-1})$
  - **Quasi-Newton:** $D^k \approx [\nabla^2 f(x^k)]^{-1}$
  - **Steepest descent:** $D^k = \boldsymbol{I}$
  - **Diagonally scaled:** D^k diagonal with $D^k_{ii} \approx \left(\frac{\partial^2 f(x^k)}{(\partial x_i)^2}\right)^{-1}$
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  - $\ldots$

Exercise: Verify that $\langle \nabla f(x^k), d^k \rangle < 0$ for above choices

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  Exercise: Prove that the latter condition ensures that $\{x^k\}$ does not converge to nonstationary points.
  Sketch: Say, $x^k \to \bar{x}$; then for sufficiently large m and n, $(m > n)$

$$x^m \approx x^n \approx \bar{x}, x^m \approx x^n - \left(\sum_{k=n}^{m-1} \eta_k\right) \nabla f(\bar{x}).$$

  The sum can be made arbitrarily large, contradicting nonstationarity of $\bar{x}$

Stepsize selection*

- **Exact:** $\eta_k := \operatorname*{argmin}_{\eta \geq 0} f(x^k + \eta d^k)$
- **Limited min:** $\eta_k = \operatorname*{argmin}_{0 \leq \eta \leq s} f(x^k + \eta d^k)$
- **Armijo-rule**. Given **fixed** scalars, s, β, σ with $0 < \beta < 1$ and $0 < \sigma < 1$ (chosen experimentally). Set

$$\eta_k = \beta^{m_k} s,$$

where we **try** $\beta^m s$ for $m = 0, 1, \ldots$ until **sufficient descent**

$$f(x^k) - f(x + \beta^m s d^k) \geq -\sigma \beta^m s \langle \nabla f(x^k), d^k \rangle$$

If $\langle \nabla f(x^k), d^k \rangle < 0$, stepsize guaranteed to exist
Usually, σ small $\in [10^{-5}, 0.1]$, while β from $1/2$ to $1/10$ depending on how confident we are about initial stepsize s.

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- Stepsize computation can be expensive
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Barzilai & Borwein stepsizes

$$x^{k+1} = x^k - \eta^k \nabla f(x^k), \quad k = 0, 1, \ldots$$

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$$x^{k+1} = x^k - \eta^k \nabla f(x^k), \quad k = 0, 1, \ldots$$

$$\eta_k = \frac{\langle u^k, v^k \rangle}{\|v^k\|^2}, \qquad \eta_k = \frac{\|u^k\|^2}{\langle u^k, v^k \rangle}$$

$$u^k = x^k - x^{k-1}, \quad v^k = \nabla f(x^k) - \nabla f(x^{k-1})$$

Challenge. Analyze convergence of GD using BB stepsizes.

Exercise

- Let D be the $(n-1) \times n$ *differencing* matrix

$$D = \begin{pmatrix} -1 & 1 & & & \\ & -1 & 1 & & \\ & & & \ddots & \\ & & & -1 & 1 \end{pmatrix} \in \mathbb{R}^{(n-1)\times n},$$

- $f(x) = \frac{1}{2}\|D^T x - b\|_2^2 = \frac{1}{2}(\|D^T x\|_2^2 + \|b\|_2^2 - 2\langle D^T x, b\rangle)$
- Notice that $\nabla f(x) = D(D^T x - b)$
- Try different choices of b, and different initial vectors x_0
- **Exercise:** Experiment to see how large n must be before gradient method starts outperforming CVX
- **Exercise:** Minimize $f(x)$ for large n; e.g., $n = 10^6$, $n = 10^7$
- **Exercise:** Repeat same exercise with constraints: $x_i \in [-1, 1]$.

Convergence

(remarks)

Gradient descent – convergence

Assumption: **Lipschitz continuous gradient**; denoted $f \in C_L^1$

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- Gradient vectors of closeby points are close to each other
- Objective function has "bounded curvature"
- Speed at which gradient varies is bounded
- **Exercise:** If $f \in C_L^1$ is twice diff. then $\|\nabla^2 f(x)\|_2 \leq L$.

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Convergence of gradient norm

Theorem. Let $f \in C_L^1$. $\|\nabla f(x^k)\|_2 \to 0$ as $k \to \infty$

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Convergence rate: function suboptimality

Theorem. Let $f \in C^1_L$ be convex; let $\{x^k\}$ be generated as above, with $\eta_k = 1/L$. Then, $f(x^{T+1}) - f(x^*) = O(1/T)$.

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Theorem. If $f \in S^1_{L,\mu}$, $\eta = \frac{2}{L+\mu}$, and $\{x^k\}$ generated by GD. Then, $f(x^T) - f^* \le \frac{L}{2}\left(\frac{\kappa-1}{\kappa+1}\right)^{2T} \|x^0 - x^*\|_2^2$, where $\kappa := L/\mu$ is the *condition number*.

Proof

(sketches)

Key result: The Descent Lemma

Lemma (Descent lemma). Let $f \in C_L^1$. Then,

$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2}\|x - y\|_2^2$$

Key result: The Descent Lemma

Lemma (Descent lemma). Let $f \in C_L^1$. Then,

$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|_2^2$$

Proof. By Taylor's theorem, for $z_t = y + t(x - y)$ we have

$$f(x) = f(y) + \int_0^1 \langle \nabla f(z_t), x - y \rangle dt.$$

Adding and subtracting $\langle \nabla f(y), x - y \rangle$ we obtain

$$\begin{aligned}
|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| &= \left| \int_0^1 \langle \nabla f(z_t) - \nabla f(y), x - y \rangle dt \right| \\
&\leq \int_0^1 |\langle \nabla f(z_t) - \nabla f(y), x - y \rangle| dt \\
&\leq \int_0^1 \|\nabla f(z_t) - \nabla f(y)\|_2 \|x - y\|_2 dt \\
&\leq L \int_0^1 t \|x - y\|_2^2 dt \\
&= \tfrac{L}{2}\|x - y\|_2^2.
\end{aligned}$$

Bounds $f(x)$ above and below with quadratic functions

Descent lemma – corollary

Coroll. 1 If $f \in C_L^1$, and $0 < \eta_k < 2/L$, then $f(x^{k+1}) < f(x^k)$

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$$\begin{aligned} f(x^{k+1}) &\leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \tfrac{L}{2}\|x^{k+1} - x^k\|_2 \\ &= f(x^k) - \eta_k \|\nabla f(x^k)\|_2^2 + \tfrac{\eta_k^2 L}{2}\|\nabla f(x^k)\|_2^2 \\ &= f(x^k) - \eta_k(1 - \tfrac{\eta_k}{2}L)\|\nabla f(x^k)\|_2^2 \end{aligned}$$

If $\eta_k < 2/L$ we have descent. min over η_k to get best bound, giving $\eta_k = 1/L$.

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Alternative bigger picture

Minimize global upper bound:

$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|^2$$

$$f(x) \leq F(x, y), \text{ where } F(x, x) = f(x)$$

Explore: Other global upper bounds and corresponding algorithms.

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$$\|\nabla f(x^k)\|_2 \to 0 \quad \text{as} \quad k \to \infty.$$

- $\min\limits_{0\leq k\leq T}\|\nabla f(x^k)\|^2 \leq \frac{1}{T+1}\sum_{k=0}^{T}\|\nabla f(x^k)\|^2$, so $O(\frac{1}{\varepsilon})$ for $\|\nabla f\|^2 \leq \varepsilon$
- Notice, we **did not require** f to be convex ...

SGD

Convergence rate – strongly convex

Theorem. If $f \in S^1_{L,\mu}$, $0 < \eta < 2/(L+\mu)$, then the gradient method generates a sequence $\{x^k\}$ that satisfies

$$\|x^k - x^*\|_2^2 \leq \left(1 - \frac{2\eta\mu L}{\mu + L}\right)^k \|x^0 - x^*\|_2.$$

Moreover, if $\eta = 2/(L+\mu)$ then

$$f(x^k) - f^* \leq \frac{L}{2}\left(\frac{\kappa - 1}{\kappa + 1}\right)^{2k} \|x^0 - x^*\|_2^2,$$

where $\kappa := L/\mu$ is the ***condition number***.

Convergence – strongly convex case

Assumption: ***Strong convexity***; denote $f \in S^1_{L,\mu}$

$$f(x) \geq f(y) + \langle \nabla f(y),\, x - y \rangle + \frac{\mu}{2}\|x - y\|_2^2$$

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Descent lemma convex corollary

Corollary 2. If f is a **convex** function $\in C^1_L$, then

$$\frac{1}{L}\|\nabla f(x) - \nabla f(y)\|_2^2 \leq \langle \nabla f(x) - \nabla f(y), x - y \rangle,$$

Exercise: Prove Cor. 2. (*Hint:* Consider $\phi(y) = f(y) - \langle \nabla f(x_0), y \rangle$).

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Valuable refinement for the strongly convex case. . .

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$$\langle \nabla f(x) - \nabla f(y),\, x - y\rangle \geq \frac{\mu L}{\mu + L}\|x - y\|_2^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|_2^2$$

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- If $\mu = L$, then immediate from strong convexity and Cor. 2

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$$\frac{1}{L}\|\nabla f(x) - \nabla f(y)\|_2^2 \leq \langle \nabla f(x) - \nabla f(y),\, x - y\rangle,$$

Thm 2. Suppose $f \in S^1_{L,\mu}$. Then, for any $x, y \in \mathbb{R}^n$

$$\langle \nabla f(x) - \nabla f(y),\, x - y\rangle \geq \frac{\mu L}{\mu + L}\|x - y\|_2^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|_2^2$$

- Consider the **convex** function $\phi(x) = f(x) - \frac{\mu}{2}\|x\|_2^2$
- If $\mu = L$, then immediate from strong convexity and Cor. 2
- If $\mu < L$, then $\phi \in C^1_{L-\mu}$; now invoke Cor. 2

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Let's put this to use now....

Convergence rate – strongly convex

Theorem. If $f \in S^1_{L,\mu}$, $0 < \eta < 2/(L+\mu)$, then the gradient method generates a sequence $\{x^k\}$ that satisfies

$$\|x^k - x^*\|_2^2 \leq \left(1 - \frac{2\eta\mu L}{\mu + L}\right)^k \|x^0 - x^*\|_2.$$

Moreover, if $\eta = 2/(L+\mu)$ then

$$f(x^k) - f^* \leq \frac{L}{2}\left(\frac{\kappa - 1}{\kappa + 1}\right)^{2k} \|x^0 - x^*\|_2^2,$$

where $\kappa := L/\mu$ is the ***condition number***.

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where we used Thm. 2 with $\nabla f(x^*) = 0$ for last inequality.

Exercise: Complete the proof of the theorem now.

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- ⋆ Want to prove the well-known $O(1/T)$ rate
- ⋆ Let $\eta_k = 1/L$
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Since $\eta_k < 2/L$, it follows that $r_{k+1} \leq r_k$

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Now we have a bound on the gradient norm...

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Recall $f(x^{k+1}) \leq f(x^k) - \frac{1}{2L}\|g^k\|_2^2$; subtracting f^* from both sides

$$\Delta_{k+1} \leq \Delta_k - \frac{\Delta_k^2}{2Lr_0^2} = \Delta_k\big(1 - \tfrac{\Delta_k}{2Lr_0^2}\big)$$

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$$\Delta_{k+1} \leq \Delta_k(1 - \beta_k) \implies \quad \frac{1}{\Delta_{k+1}} \geq \frac{1}{\Delta_k}(1 + \beta_k) = \frac{1}{\Delta_k} + \frac{1}{2Lr_0^2}.$$

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- Sum both sides over $k = 0, \ldots, T$ (telescoping) to obtain

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- Use descent lemma to bound $\Delta_0 \leq (L/2)\|x^0 - x^*\|_2^2$; simplify

$$f(x^T) - f(x^*) \leq \frac{2L\Delta_0\|x^0 - x^*\|_2^2}{T+4} = O(1/T).$$

SGD

$$x \leftarrow x - \eta g$$

Why SGD?

Regularized Empirical Risk Minimization

$$\min_x \quad \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, x^T a_i) + \lambda r(x).$$

(e.g., logistic regression, deep learning, SVMs, etc.)

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- training data: $(a_i, y_i) \in \mathbb{R}^d \times \mathcal{Y}$ (i.i.d.)
- large-scale ML: Both d and n are large:
  - d: dimension of each input sample
  - n: number of training data points / samples
- Assume training data "sparse"; so total datasize $\ll dn$.
- Running time $O(\#\text{nnz})$

Finite-sum problems

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Gradient / subgradient methods

$$\begin{aligned} x_{k+1} &= x_k - \eta_k \nabla f(x_k) \\ x_{k+1} &= x_k - \eta_k g(x_k), \quad g \in \partial f(x_k) \end{aligned}$$

If n is large, each iteration above is expensive

Stochastic gradients

At iteration k, we randomly pick an integer
$i(k) \in \{1, 2, \ldots, n\}$

$$x_{k+1} = x_k - \eta_k \nabla f_{i(k)}(x_k)$$

- The update requires only gradient for $f_{i(k)}$
- Uses unbiased estimate $\mathbb{E}[\nabla f_{i(k)}] = \nabla f$
- One iteration now n times faster using $\nabla f(x)$
- Can such a method work? If so, how fast? Why?

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- Assume all variables involved are **scalars**.

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- Notice now that

$$x^* \in [\min_i x_i^*, \max_i x_i^*] =: R$$

(Use: $\sum_i a_i b_i = \sum_i a_i^2 (b_i/a_i)$)

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- $\nabla f_i(x)$ has **same sign** as $\nabla f(x)$. So using $\nabla f_i(x)$ **instead** of $\nabla f(x)$ also ensures progress.

Intuition – (Bertsekas)

- Assume all variables involved are **scalars**.

$$\min \quad f(x) = \tfrac{1}{2}\sum\nolimits_{i=1}^{n}(a_i x - b_i)^2$$

- Notice: $x^* \in \left[\min_i x_i^*, \max_i x_i^*\right] =: R$
- If we have a scalar x that lies outside R?
- We see that

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$$\nabla f(x) = \sum\nolimits_i a_i(a_i x - b_i)$$

- $\nabla f_i(x)$ has **same sign** as $\nabla f(x)$. So using $\nabla f_i(x)$ **instead** of $\nabla f(x)$ also ensures progress.
- But once inside region R, **no guarantee** that SGD will make progress towards optimum.

SGD: two variants

$$\min \frac{1}{n} \sum_i f_i(x)$$

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Explore. Which version would you use? Why?

SGD: mini-batches

$$\min_x \quad f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$$

Idea: Use a mini-batch of stochastic gradients

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- Iteration k samples set I_k, and uses $|I_k|$ stochastic gradients
- Increases parallelism, reduces communication

Explore: Large mini-batches not that "favorable" for DNNs.
(also known as: "large-batch training")

SGD: some theoretical challenges

$$x_{k+1} = x_k - \eta_k \nabla f_{i(k)}(x_k)$$

- Proving that it "works"
- Theoretical results lagging behind practice (without replacement SGD widely used, most published theory studies with replacement)

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Explore: Why does SGD work so well for neural networks? (i.e., why does it deliver such low training losses despite non-convexiity, and how does it influence generalization behavior of neural networks?)

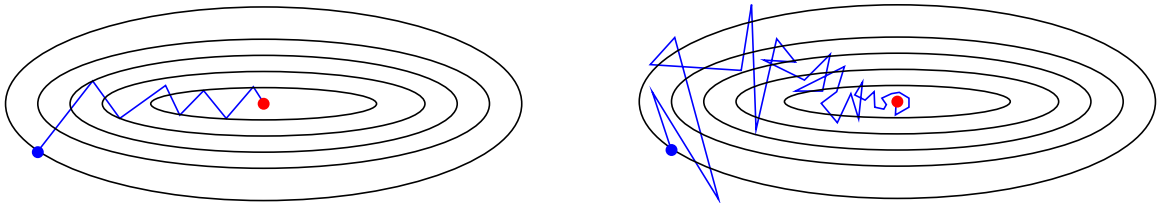
SGD for empirical risk / finite sums

$$\min_{x \in \mathbb{R}^d} f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x)$$

- **Iteration**: $x^{k+1} = x^k - \eta_k f'_{i(t)}(x^k)$
  - Sampling with replacement: $i(k) \sim \text{Unif}(\{1, \ldots, n\})$
  - Polyak-Ruppert averaging: $\bar{x}_k = \frac{1}{k+1} \sum_{j=0}^{k} x^j$
- **Convergence rate** if each f_i convex L-smooth, and f is μ-strongly-convex:

$$\mathbb{E}[f(\bar{x}_k) - f(x^*)] \leqslant \begin{cases} O(1/\sqrt{k}) & \text{if } \eta_k = 1/(L\sqrt{k}) \\ O(L/(\mu k)) = O(\kappa/k) & \text{if } \eta_k = 1/(\mu k) \end{cases}$$

SGD vs GD (strongly convex case)



GD SGD

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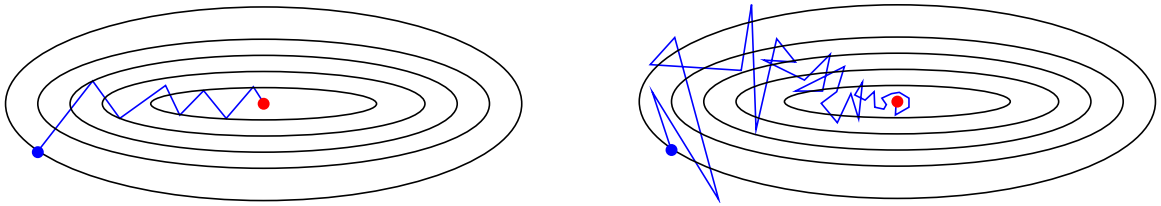

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  – Linear (e.g., exponential) convergence rate in $O(e^{-k/\kappa})$
  – Iteration complexity is linear in n ($O(n \log 1/\varepsilon)$)

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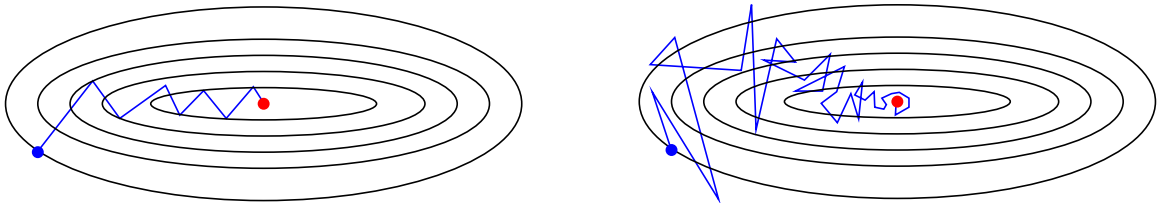

GD SGD

- Batch GD:
  – Linear (e.g., exponential) convergence rate in $O(e^{-k/\kappa})$
  – Iteration complexity is linear in n ($O(n \log 1/\varepsilon)$)
- SGD:
  – Sampling with replacement: $i(k)$ random element of $\{1, \ldots, n\}$
  – Convergence rate $O(\kappa/k)$
  – Iteration complexity independent of n ($O(1/\varepsilon^2)$)